How to create a travel router with Raspberry Pi + OpenWRT

Credit: https://www.youtube.com/watch?v=jlHWnKVpygw&t=364s

Things to Prepare:

- Ethernet port
- Raspberry Pi 2 and above for better performance
- Micro SD (minimum 8GB) and micro SD adapter
- Power Supply
- 1 laptop/PC
- WiFi Adapter (must be compatible with OpenWRT)

Note: The Raspberry Pi needs 2 WiFi interfaces to work as a travel router — the built-in WiFi acts as the receiver (connects to existing internet) and the USB WiFi Adapter acts as the emitter (broadcasts to your devices). If your Raspberry Pi has no built-in WiFi, you will need 2 USB WiFi adapters.

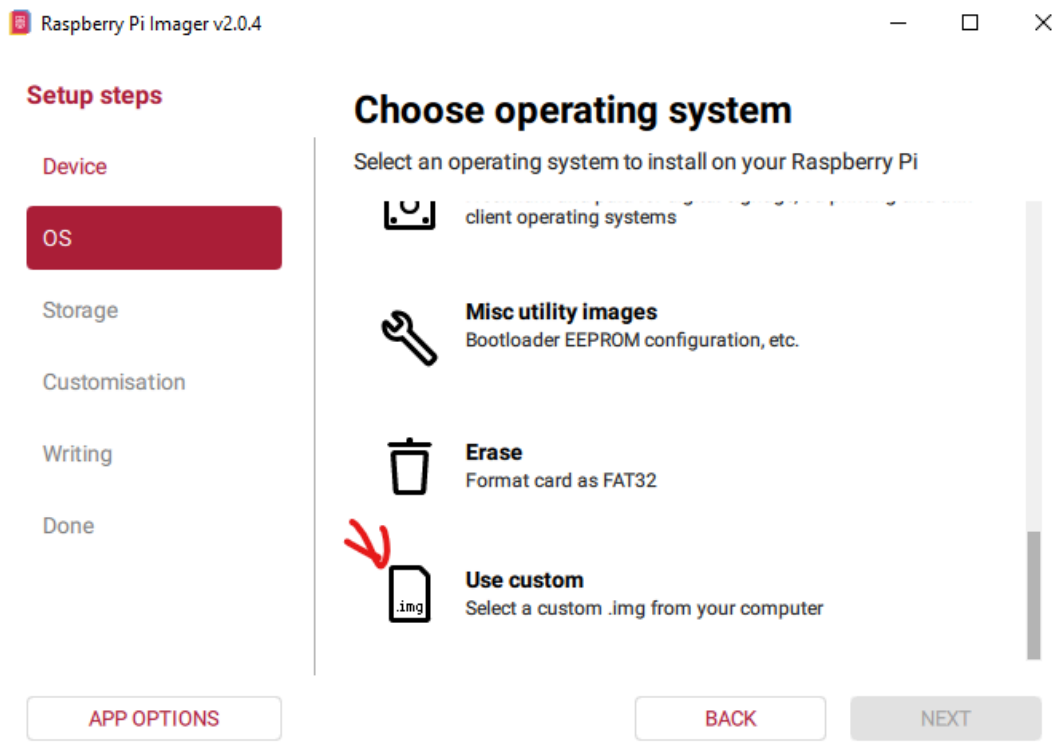
Step 1: Download OpenWrt on Raspberry Pi 5

- Find this link: https://openwrt.org/toh/raspberry_pi_foundation/raspberry_pi
- Choose the Open firmware that match Raspberry Pi 5

Installation

Model	Raspberry Pi	Raspberry Pi	Raspberry Pi	Raspberry Pi	Raspberry Pi	Raspberry Pi	Raspberry Pi 2	Raspberry Pi 2	Raspberry Pi 3	Raspberry Pi 3	Raspberry Pi 3	Raspberry Pi 3	Raspberry Pi 4	Raspberry Pi 4	Raspberry Pi 4	Raspberry Pi 5	Raspberry Pi 5	Raspberry Pi Zero
Version	500	A	B	B+	Zero	Zero W	B 1.0/1.1	B 1.2	A+	B	B+	Compute Module 3	400	B	Compute Module 4		Compute Module 5	2W
Supported Current Rel	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4	24.10.4
Firmware OpenWrt Install URL	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image	Factory image
Firmware OpenWrt Upgrade URL	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image	Sysupgrade image

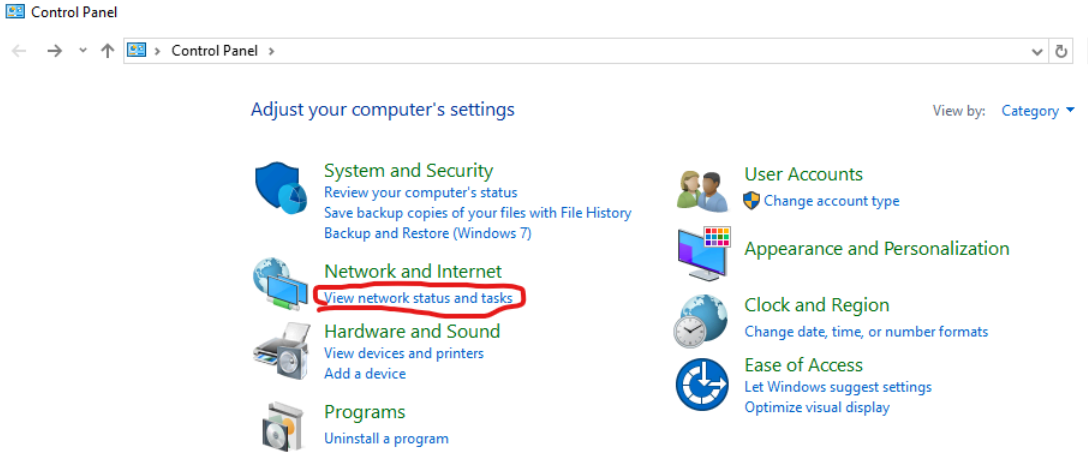
- After download the firmware file, use Raspberry Pi Imager to install OpenWRT firmware with “Use Custom”

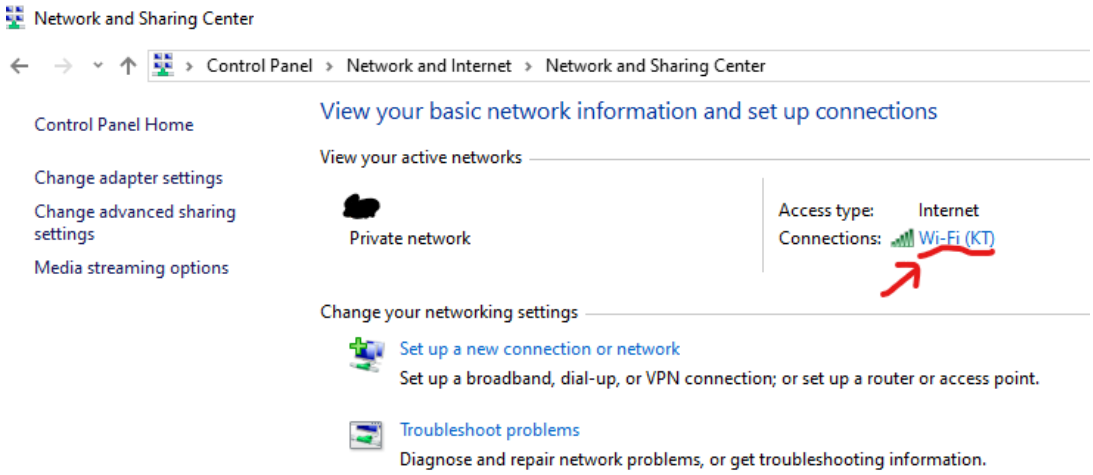


- Then after installing the firmware on Micro SD, plug it to Raspberry Pi

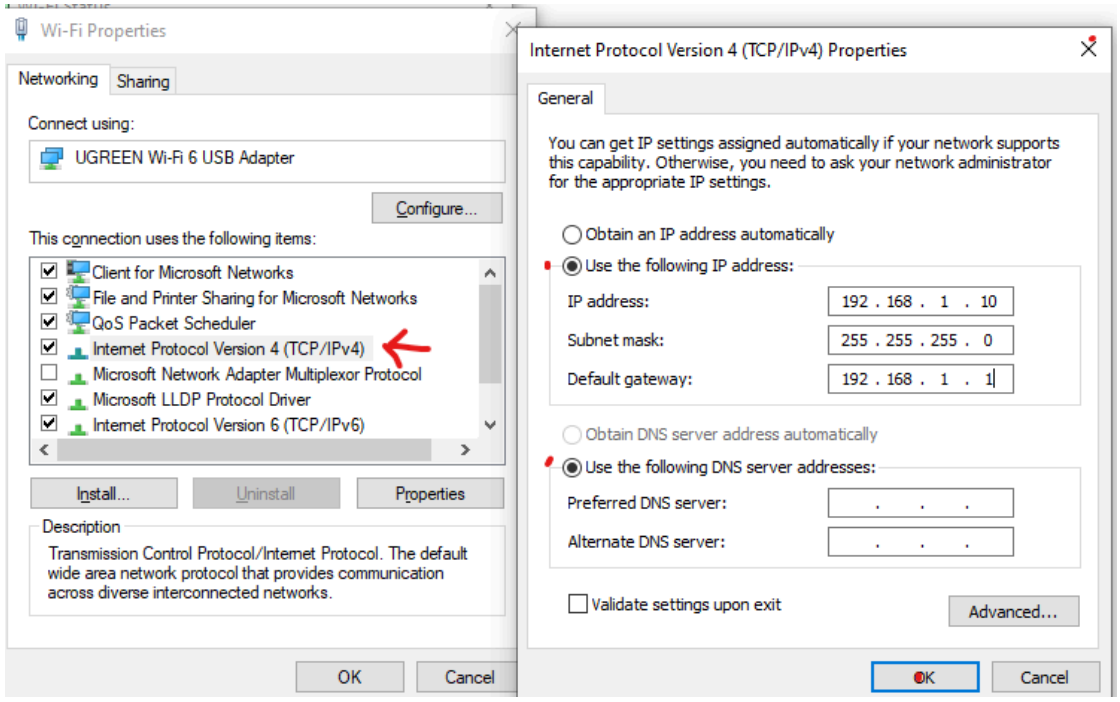
Step 2: Change the IP Address from Laptop/PC to static IP for example: 192.168.1.10

- Search Control Panel to find **Network and Internet**





- Look for Property -> click on Internet Protocol Version 4 (TCP/IP) to change IP



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Step 3: Open Command Prompt/Terminal

- At the moment the Raspberry Pi is supposed to have an IP address 192.168.1.1

- Open Terminal and type: **ssh root@192.168.1.1** and type **yes**

```
C:\Users\chuck>ssh root@192.168.1.1
The authenticity of host '192.168.1.1 (192.168.1.1)' can't be established.
ED25519 key fingerprint is SHA256:+UcSqY1qr1GF1UH2Nyxy949dNUO3jSTDLSTqfeqjL64.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.1' (ED25519) to the list of known hosts.


BusyBox v1.33.1 (2021-07-31 17:21:01 UTC) built-in shell (ash)


  _______                     ________        __
 |       |.-----.-----.-----.|  |  |  |.----.|  |_
 |   -   ||  _  |  -__|     ||  |  |  ||   _||   _|
 |_______||   __|_____|__|__||________||__|  |____|
          |__| W I R E L E S S   F R E E D O M
 -----------------------------------------------------
 OpenWrt 21.02.0-rc4, r16256-2d5ee43dc6
 -----------------------------------------------------
=== WARNING! =====================================
There is no root password defined on this device!
Use the "passwd" command to set up a new password
in order to prevent unauthorized SSH logins.
--------------------------------------------------
root@OpenWrt:~#
```

- After that create a new password with command line: **passwd** and choose the password you can remember
- Type: **cd /etc/config**
- After that type: **ls** (to see all the folder contained in /etc/config
- Now back up the important files in case we want to reimage in the OpenWRT and start over
- There are three important files to back up: firewall, wireless, network
    - Use these command line to back up:
        - For firewall, type **cp firewall** **firewall.bk**
        - For wireless, type **cp wireless** **wireless.bk**
        - For network, type **cp network** **network.bk**
- Now to configure the important files for OpenWRT to work (in order to edit the file, make sure to click i on keyboard to make it work)
- To configure network, type: vi network
    - In the config interface 'lan', replace IP address in **option ipaddr** to any private network you like: example From 192.168.1.1 to 10.172.16.1
    - Add one more line under the line **option ip6assign '60'**, type: **option force_link 1** ,this means to make a static IP specifically for this Raspberry Pi, so no one else can have this IP
    - Create two more config interface "wwan" and "vpnclient"

```
config interface 'lan'
        option device 'br-lan'
        option proto 'static'
        option ipaddr '10.71.71.1'
        option netmask '255.255.255.0'
        option ip6assign '60'
        option force_link '1'

config interface 'wwan'
        option proto 'dhcp'
        option peerdns '0'
        option dns '1.1.1.1 8.8.8.8'

config interface 'vpnclient'
        option ifname 'tun0'
        option proto 'none'
```

- To save the file: Click ESC button -> type :wq
- To configure firewall, type: vi firewall
    - In the config zone, replace REJECT to ACCEPT

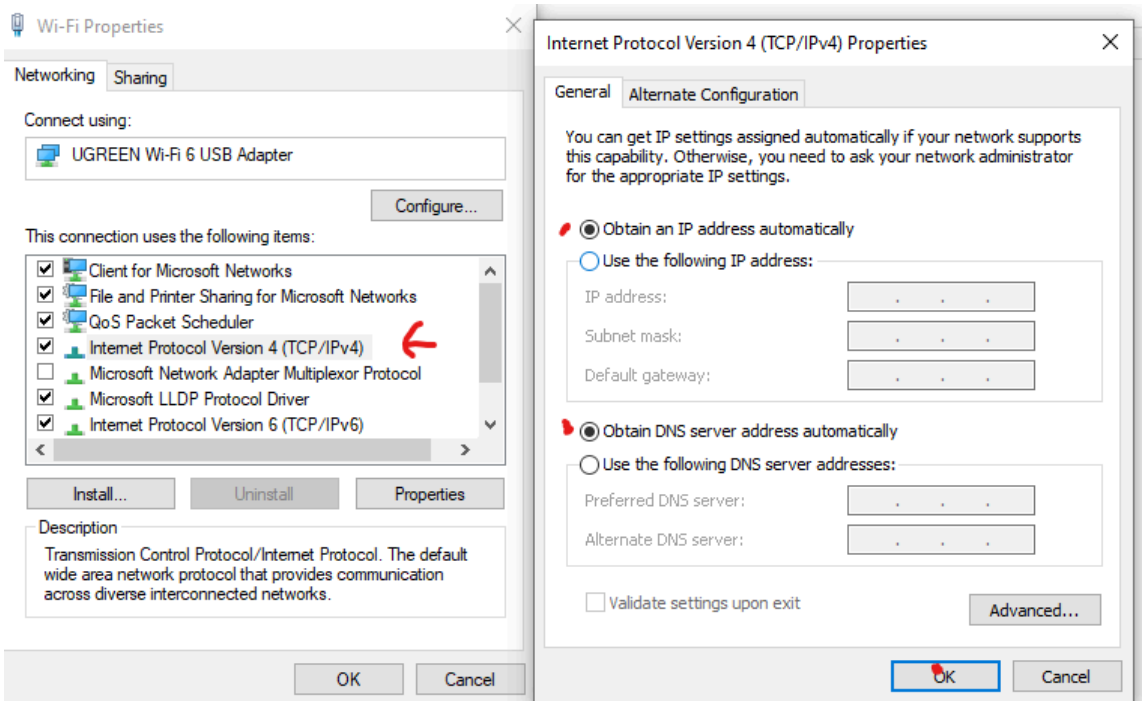
```
config defaults
        option syn_flood        1
        option input            ACCEPT
        option output           ACCEPT
        option forward          REJECT
# Uncomment this line to disable ipv6 rules
#       option disable_ipv6     1

config zone
        option name             lan
        list   network          'lan'
        option input            ACCEPT
        option output           ACCEPT
        option forward          ACCEPT

config zone
        option name             wan
        list   network          'wan'
        list   network          'wan6'
        option input            REJECT
        option output           ACCEPT
        option forward          REJECT
        option masq             1
        option mtu_fix          1

config forwarding
        option src              lan
        option dest             wan
- firewall 1/208 0%
```

- To save the file: Click ESC button -> type :wq

Now reboot the Raspberry Pi by using command: **reboot**

Step 4: Put the laptop/PC back to automatic IP

- Open Control Panel → Network and Internet → Network and Sharing Center
- Click Change adapter settings → right-click your network adapter → Properties
- Click Internet Protocol Version 4 (TCP/IPv4) → Properties
- Select "Obtain an IP address automatically"
- Select "Obtain DNS server address automatically"
- Click OK to save



Step 5: log back in but now with your IP address

- **ssh root@[your_private_ip_address] . For example: ssh root@10.172.16.1**
- Make change in wireless configuration
  - To configure wireless, type: vi /etc/config/wireless
  - Replace this:

```
config wifi-device 'radio0'
        option type 'mac80211'
        option channel '36'
        option hwmode '11a'
        option path 'platform/soc/fe300000.mmcnr/mmc_host/mmc1/mmc1:0001/mmc1:0001:1'
        option htmode 'VHT80'
        option disabled '1'

config wifi-iface 'default_radio0'
        option device 'radio0'
        option network 'lan'
        option mode 'ap'
        option ssid 'OpenWrt'
        option encryption 'none'
```

  - To this

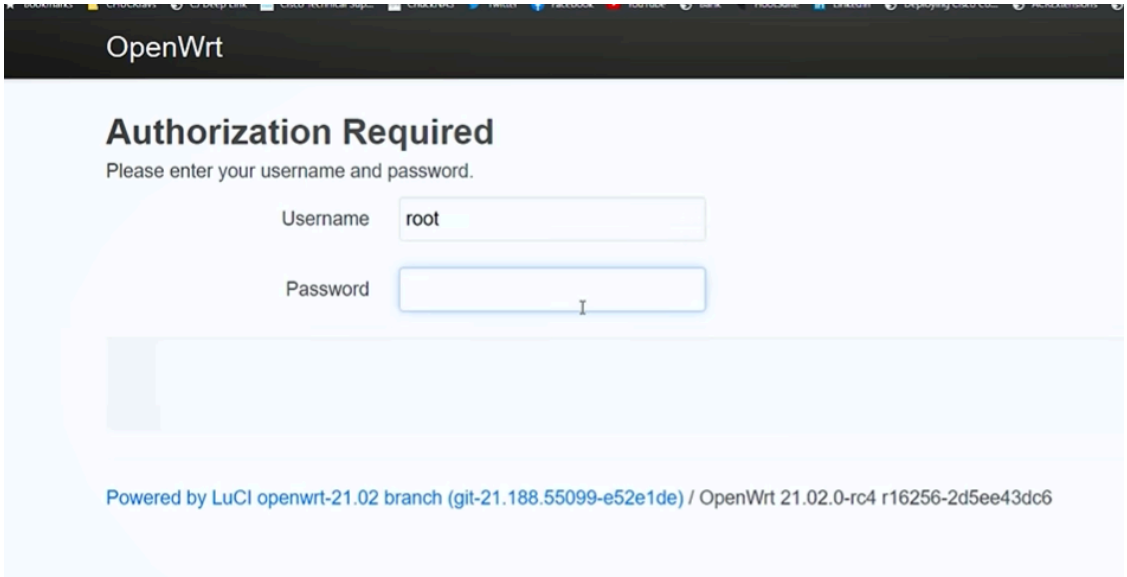
```
config wifi-device 'radio0'
        option type 'mac80211'
        option channel '7'
        option hwmode '11g'
        option path 'platform/soc/fe300000.mmcnr/mmc_host/mmc1/mmc1:0001/mmc1:0001:1'
        option htmode 'HT20'
        option disabled '0'
        option short_gi_40 '0'

config wifi-iface 'default_radio0'
        option device 'radio0'
        option network 'lan'
        option mode 'ap'
        option ssid 'OpenWrt'
        option encryption 'none'
```

- Make sure to change option disabled '1' -> '0' so that the wifi receiver is enabled to receive connection from other internet
- To save the file: Click ESC button -> type :wq

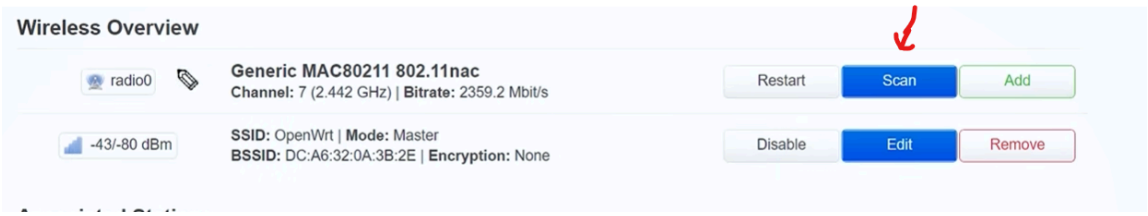
- After that use command: **uci commit wireless** to configure wireless to the interface and type: **wifi**

Step 6: Allowing your Raspberry Pi to access your internet

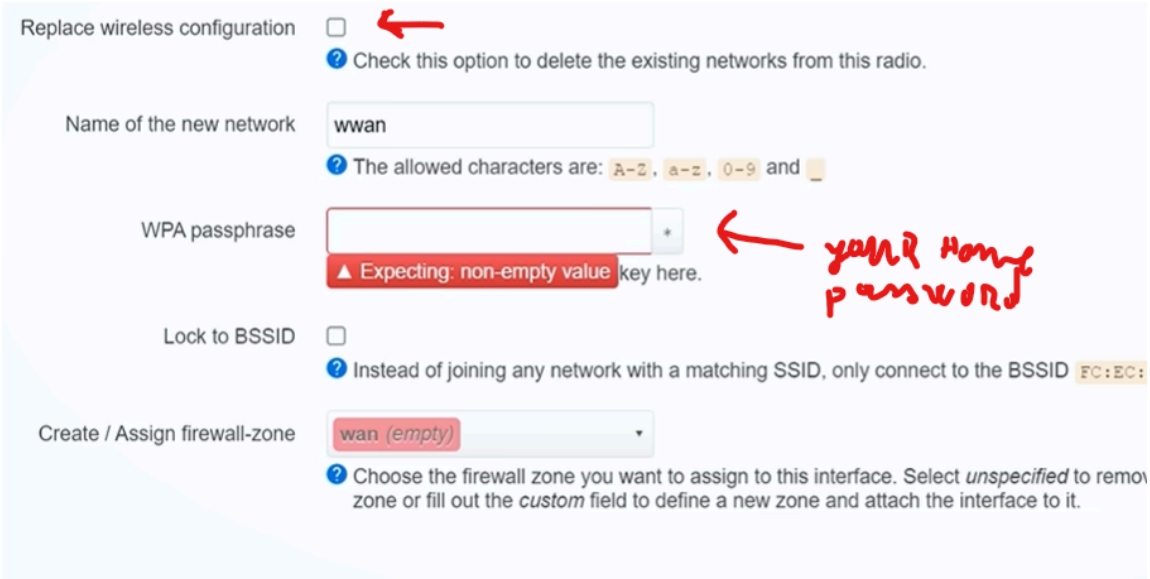
- Now open any web browser and type your private ip on web search, after that you will see this page and type your password earlier to access



- Next, go to Network -> Wireless -> Scan to find your internet



- After finding your wifi, checkmark the Replace Wireless Configuration and type in your router password and then click Submit and Save



- To test if the router work, try to ping the location using command line : **ping** google.com to see if there is good connection between router without packet dropped

Step 7: Make the Wifi adapter to be recognized by Raspberry Pi and acts an a wifi emitter

- To make the Raspberry Pi recognize the Wifi Adapter, we must install driver on Raspberry Pi
- So first update the OpenWRT, type: **opkg update**
- (If it says failed to download, try to reboot by typing: reboot)
- Next install packages for OpenWRT , type: **opkg install kmod-rt2800-lib kmod-rt2800-usb kmod-rt2x00-lib kmod-rt2x00-usb kmod-usb-core kmod-usb-uhci kmod-usb-ohci kmod-usb2 usbutils openvpn-openssl luci-app-openvpn nano**
- Plugging the wifi adapter, type to see if the Raspberry Pi recognize it: **lsusb**
- To test is the driver works with the wifi adapter try: ifconfig wlan1 up
- (if not say not recognized, try: ifconfig phy-ap1 up)

Step 8: Once again configure wireless file, type: vi /etc/config/wireless

- In the Config Wifi-device ‘radio1’

```
config wifi-device 'radio1'
        option type 'mac80211'
        option channel '11'
        option hwmode '11g'
        option path 'scb/fd500000.pcie/pci0000:00/0000:00:00.0/0000:01:00.0/
        option htmode 'HT20'
        option disabled '1'

config wifi-iface 'default_radio1'
        option device 'radio1'
        option network 'lan'
        option mode 'ap'
        option ssid 'OpenWrt'
        option encryption 'none'
```

- Make sure to change **Option disabled ‘1’** -> **Option disabled ‘0’** in order to enable the Wifi Adapter to emit the network for your device to connect
- In the Config Wifi-device ‘default_radio1’
    - In the option ssid, you can replace OpenWrt to any name you like because it will be the name for your network, such as option ssid ‘OpenWrt_router’
    - In the option encryption, change ‘none’ to ‘psk2’
    - Add one more line under the option encryption, type: option key ‘password’, as this will be the password you will use for your wifi
    - To save the file: Click ESC button -> type :wq

To verify everything is working correctly:
- Connect a device to the WiFi network you just created
- Open Terminal or Command Prompt and ping google.com
- If you see replies with no packet loss, the travel router is working correctly

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Troubleshooting: Factory Reset

If something goes wrong and you are not sure how to fix it, the easiest solution is to reset OpenWRT to factory defaults and start over.

Step 1: Reset OpenWRT to factory defaults:

```
firstboot -y && reboot
```

Step 2: Clear the old SSH keys on your laptop/PC:

```
ssh-keygen -R {your_private_ip}
ssh-keygen -R 192.168.1.1
```

Step 3: In Control Panel, set your laptop/PC IPv4 back to static IP

(192.168.1.10) with default gateway left blank — same as Step 2 of this guide.

Step 4: SSH back in and start over:

```
ssh root@192.168.1.1
```